



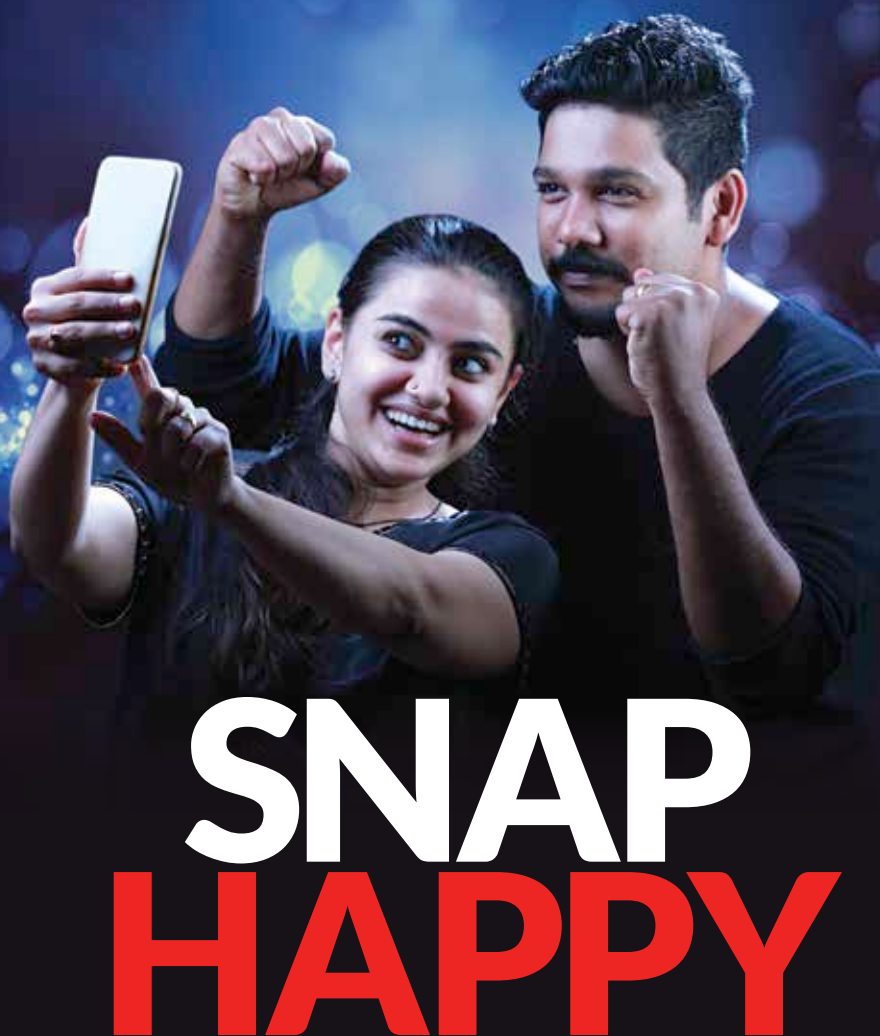
Netflix-like Game Streaming by Google

Google may unveil its Netflix-esque game streaming service in the Game Developer's Conference keynote to be held on March 19. <https://digit.in/mar19-34>



HoloLens app on Android

Microsoft's Dynamics 365 Remote Assist tool will be available on Android as a preview feature by April. <https://digit.in/mar19-35>



Smartphone cameras are all about the software rather than the hardware these days

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Over the past ten years, the cameras in smartphones are the components that have evolved the most.

Except for a few outliers, most of the image sensors come from two companies – Sony and Samsung. We have seen back side illumination (BSI) sensors getting cheaper, improved pixel arrangement with the quad bayer mosaic for low light conditions, faster focusing using phase

detection, and an increasing number of lens options for the primary cameras. Powerful processors support the sensors by providing capabilities such as 4K video recording. Neuromorphic hardware, or chips that imitate the structure of biological brains, combined with AI technologies further process the image in various ways. While different models of smartphones might use the same processor, the differentiating factor in the higher end devices is often the capabilities of the camera. All of this hardware innovation is supported by the software, which now works before, during and after a click to make sure that the user has a versatile and powerful camera in their hands.

HDR

The primary camera on the Honor View 20 shoots staggering 48 MP

photos. One of the resolution options available is called AI Ultra Clarity, which uses techniques similar to the ones used to produce HDR images. Multiple exposures are captured at a single shutter trigger, after which the two neural processing units on the Kirin 980 processor analyse the images. The overexposed portions are minimised, the colour reproduction is optimised, and the details in the shadows are restored, by synthesizing different parts of the multiple exposures into a single photo. The AI Ultra Clarity mode is available as a separate setting under camera resolutions. In low light conditions, the phone determines if the camera is being held in a hand by using signals from the gravity sensor and the gyroscope. The duration of the exposure is adjusted depending on these signals, with multiple images captured over an exposure that lasts between two to six seconds, if the camera is not on a tripod. These images are processed by AI to widen the dynamic range in low light shots, as well as reduce noise and bring out details.

The Google Pixel 3 has just about the best camera for taking low light shots. This is because of a feature called Night Sight, which improves on the HDR+ capabilities available through the Google Camera app on Pixel and Nexus devices. The app has been unofficially ported for other devices. HDR+ has what Google calls zero shutter lag, as it continuously keeps capturing images, and uses up to nine images captured slightly before and after the shutter is triggered, to synthesize an image. Night Sight, which was rolled out as a software update to all Pixel devices after the Pixel 3 was launched, works by taking multiple exposures in quick succession to create well lit, sharp images with minimal noise, even in low light conditions. The feature allows Pixel users to use natural light sources, and not depend on the flash to illuminate a subject. Instead of a zero shutter lag, Night Sight uses a positive shutter lag, that prompts you to hold the camera still for up to six seconds after the image has been clicked. Up to fifteen images